Step 1: choose criteria and weights

| | |
|---|---|
| must last 10 days | 0.1 |
| water the plants | 0.1 |
| automatic | 0.3 |
| portable | 0.1 |
| connects to rain barrel | 0.3 |
| self sufficent | 0.1 |
| sum (should = 1) | 1 |

Step 2: use criteria and weights to compare and select a design (use a weighted decision matrix)

**All members must use same criteria and weights and the same design options in the below weighted decision matrices**
**vote (e.g. use - 1, 0, 1 scale)**

| Sammi | Weight as decimal | drip | sprinkler | water fountain | surface | subsurface | flood |
|---|---|---|---|---|---|---|---|
| must last 10 days | 0.1 | 1 | 1 | 1 | 1 | 1 | 1 |
| water the plants | 0.1 | 1 | 1 | 1 | 1 | 1 | 1 |
| automatic | 0.3 | 1 | 1 | 0 | 1 | 1 | 1 |
| portable | 0.1 | 1 | 1 | 1 | 1 | 1 | 1 |
| connects to rain barrel | 0.3 | 1 | 0 | 0 | 1 | 1 | 1 |
| self sufficent | 0.1 | 0 | 0 | 1 | 1 | 1 | 1 |
| | total score | 0.9 | 0.6 | 0.4 | 1 | 1 | 1 |

| Lauren | Weight as decimal | drip | sprinkler | water fountain | surface | subsurface | flood |
|---|---|---|---|---|---|---|---|
| must last 10 days | 0.1 | 1 | 1 | 0 | 1 | 1 | 1 |
| water the plants | 0.1 | 1 | 1 | 1 | 1 | 1 | 1 |
| automatic | 0.3 | 1 | 1 | 0 | 1 | 1 | 1 |
| portable | 0.1 | 0 | 1 | 1 | 1 | 0 | 1 |
| connects to rain barrel | 0.3 | 1 | 1 | 1 | 1 | 1 | 1 |
| self sufficent | 0.1 | 1 | 1 | 1 | 1 | 1 | 1 |
| | total score | 0.9 | 1 | 0.6 | 1 | 0.9 | 1 |

| Davis | Weight as decimal | drip | sprinkler | water fountain | surface | subsurface | flood |
|---|---|---|---|---|---|---|---|
| must last 10 days | 0.1 | 1 | 0 | -1 | 1 | 1 | -1 |
| water the plants | 0.1 | 1 | 1 | 1 | 1 | 1 | 1 |
| automatic | 0.3 | 1 | 1 | -1 | | | |
| portable | 0.1 | 0 | 1 | -1 | -1 | -1 | 1 |
| connects to rain barrel | 0.3 | 1 | -1 | -1 | 1 | 1 | -1 |
| self sufficent | 0.1 | 1 | 1 | 0 | 1 | 1 | 0 |
| | total score | 0.9 | 0.3 | -0.7 | 0.5 | 0.5 | -0.2 |

| Peter Schwindt | Weight as decimal | drip | sprinkler | water fountain | surface | subsurface | flood |
|---|---|---|---|---|---|---|---|
| must last 10 days | 0.1 | 1 | 0 | -1 | 1 | 1 | -1 |
| water the plants | 0.1 | 1 | 1 | 0 | 1 | 1 | 1 |
| automatic | 0.3 | 0 | 1 | -1 | | | |
| portable | 0.1 | 1 | 0 | -1 | -1 | -1 | 0 |
| connects to rain barrel | 0.3 | 1 | -1 | -1 | 0 | 1 | -1 |
| self sufficent | 0.1 | 1 | 1 | 1 | 1 | 1 | 0 |
| | total score | 0.7 | 0.2 | -0.7 | 0.2 | 0.5 | -0.3 |

| Aidan | Weight as decimal | drip | sprinkler | water fountain | surface | subsurface | flood |
|---|---|---|---|---|---|---|---|
| must last 10 days | 0.1 | 1 | 0 | -1 | 1 | 1 | 0 |
| water the plants | 0.1 | 1 | 1 | 0 | 1 | 0 | |
| automatic | 0.3 | 0 | 1 | -1 | 1 | 1 | -1 |
| portable | 0.1 | 1 | 0 | -1 | -1 | -1 | -1 |
| connects to rain barrel | 0.3 | 1 | 0 | 1 | 0 | 1 | -1 |
| self sufficent | 0.1 | 1 | 1 | | 1 | 1 | 1 |
| | total score | 0.7 | 0.5 | -0.2 | 0.5 | 0.7 | -0.6 |